Letter Sequences & Alphabet Code (Set B)

Year 4 · **Lesson Worksheet**

Worked Examples

Example 1 — Letter Sequence: What comes next in this sequence? B, D, F, H, __

Answer: J — each letter skips one letter in the alphabet (+2 each time): B(2), D(4), F(6), H(8), J(10).

Example 2 — Position Code: If A=1, B=2, C=3 and so on, what number represents the letter K?

Answer: 11 — K is the 11th letter of the alphabet.

Example 3 — Shift Code: In a code where every letter is shifted forward by 3 places (A→D, B→E, C→F...), what does the letter M become?

Answer: P — M is shifted 3 places forward: M, N, O, P.

Spotting the Pattern (15 questions)

1. What comes next? A, C, E, G, __
2. What comes next? Z, Y, X, W, __
3. If A=1, B=2, C=3..., what number stands for the letter F? A) 5 B) 6 C) 7 D) 8 E) 9
4. Select ALL that apply: which of these letter pairs are exactly 2 letters apart in the alphabet (e.g. B and D)? A) C and E B) G and I C) M and O D) P and R E) T and U
5. What comes next in this sequence? B, E, H, K, __
6. In a code where A=1, B=2, C=3..., which letter stands for the number 15? A) M B) N C) O D) L E) K
7. What comes next? Y, W, U, S, __
8. Fill in the missing letter: A, _, E, G, I (skipping one letter each time)
9. If every letter is shifted forward by 1 place (A→B, B→C, C→D...), what does D become? A) C B) E C) F D) G E) B
10. Select ALL that apply: in the sequence C, F, I, L, O, which of these statements are true? A) The gap between each letter is always 3. B) The next letter after O would be R. C) The sequence goes backwards through the alphabet. D) The first letter is C. E) The letter after L is M.
11. What comes next? D, F, H, J, __
12. In the alphabet-position code (A=1, B=2, ...), what number represents the letter T? A) 18 B) 19 C) 20 D) 21 E) 22
13. What comes next? Q, O, M, K, __
14. If letters are shifted BACK by 2 places (C→A, D→B, E→C...), what does H become? A) E B) F C) G D) I E) J
15. Select ALL that apply: which of these are letters that come AFTER M in the alphabet? A) P B) J C) R D) L E) W

Applying the Pattern (15 questions)

1. What comes next in this sequence? B, D, G, I, L, __ (the gap alternates +2, +3, +2, +3...)
2. A code shifts every letter forward by 4 places (A→E, B→F...). What does the word "CAB" become?

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3. What comes next? A, D, H, M, __ (the gap increases by 1 each time: +3, +4, +5...)

4. In a code, A=1, B=2, C=3... Two numbers are added together and converted back to a letter: if the numbers are 7 and 8, which letter results? A) N B) O C) P D) M E) Q

5. Select ALL that apply: which sequences below correctly continue with a GAP of 4 between each letter? A) A, E, I, M, Q B) B, F, J, N, R C) C, F, I, L, O D) D, H, L, P, T E) A, D, G, J, M

6. What comes next? Z, W, T, Q, __ (the gap is always -3)

7. A code shifts every letter BACK by 3 places (D→A, E→B, F→C...). What does the word "GIF" decode to?

8. What comes next in this sequence, which alternates two interleaved patterns? A, Z, B, Y, C, __

9. In the position code (A=1...Z=26), what letter corresponds to the number you get from 26 − 6? A) S B) T C) U D) V E) W

10. Select ALL that apply: in a +5 shift code (A→F, B→G, C→H...), which of these letter pairs are correctly matched? A) A→F B) C→H C) J→O D) M→R E) Z→E (wrapping to the start of the alphabet)

11. What comes next? C, F, J, O, __ (the gap increases by 1 each time: +3, +4, +5...)

12. A code uses the rule "position number + 3, then convert back to a letter". What does the letter W become? (wrap round to A after Z if needed)

13. What comes next? L, I, F, C, __

14. Select ALL that apply: which of these statements about the sequence E, H, K, N, Q are correct? A) Each gap is +3. B) The sequence is decreasing. C) The next letter after Q is T. D) The first letter is E. E) The gap between K and N is 2.

15. In a shift code where every letter moves forward by 7 places, what does the letter S become? (wrap round to the start of the alphabet if you go past Z)

Codes in Context (15 questions)

1. What comes next in this sequence: B, E, H, K, ?

2. A code shifts every letter forward by 4 places (A→E, B→F...). What does the word "BIT" become?

3. What comes next in this sequence: Z, W, T, Q, ? A) M B) N C) O D) P E) R

4. In a position code (A=1...Z=26), what is the sum of the position numbers of the letters in the word "CAT"?

5. A puzzle gives you this pair as a worked example of a code: BAT → DCV. Work out the shift being used, then use it to encode the word "PEN".

6. In a shift code, the letter E becomes H. Using the same shift, what would the letter M become? A) O B) P C) Q D) R E) S

7. What comes next in this sequence: C, G, K, O, ?

8. A puzzle gives you this pair as a worked example of a code: HEN → KHQ. Work out the shift, then decode the word "GRJ" using the same shift.

9. In a position code (A=1...Z=26), what is the sum of the position numbers of the letters in the word "DOG"?

10. Select ALL that apply: for the sequence D, H, L, P, which of these are true? A) The gap between each letter is 4. B) The next letter would be T. C) The sequence decreases through the alphabet. D) The gap between H and L is 4. E) The first letter is D.

11. What comes next in this sequence: Y, V, S, P, ?

12. A code shifts every letter back by 3 places (D→A, E→B, F→C...). What does the word "OLW" decode to?

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13. A puzzle gives you this pair as a worked example of a code: FOX → HQZ. Work out the shift, then encode the word "CAT" using the same shift. A) ECV B) EDV C) DCV D) ECW E) EBV

14. What comes next in this sequence: E, I, M, Q, ?

15. In a position code (A=1…Z=26), what is the sum of the position numbers of the letters in the word "SUN"? A) 51 B) 52 C) 53 D) 54 E) 55

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